

Major Project Report

For

MULTI-AGENT DEVOPS INCIDENT MANAGEMENT AND RESPONSE SYSTEM

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Prepared by

Specialization	SAP ID	Name
Cloud Computing / DevOps	500124131	Atharva Sharma
Cloud Computing / DevOps	500122613	Saurabh Nautiyal
Cloud Computing / DevOps	500125209	Nikhil Kumar Singh
Cloud Computing / DevOps	500125722	Kunal Pal

School of Computer Science
UNIVERSITY OF PETROLEUM & ENERGY STUDIES
DEHRADUN – 248007, UTTARAKHAND

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REVISION HISTORY

Date	Change	Reason for Changes	Mentor Signature
26-08-2026	Initial project report draft	Initial project definition, problem statement, architecture, requirements and research direction.	
26-08-2026	Architecture revision	Added multi-agent workflow, human-in-the-loop controls, evaluation model and UML diagrams.	

Document status: Project report / synopsis draft. The report defines the proposed prototype architecture and evaluation plan; implementation details may be refined during development.

1 INTRODUCTION

1.1 Purpose of the Project

Modern DevOps environments continuously generate operational telemetry from cloud infrastructure, applications, databases, APIs and deployment pipelines. When an incident occurs, engineers must interpret this telemetry under time pressure, correlate independent signals, determine the likely root cause, select remediation actions and document the outcome. The problem becomes more difficult as systems become distributed and the number of services, dependencies and monitoring signals increases.

The proposed project introduces a multi-agent incident-management layer in which specialised AI agents collaborate across the incident lifecycle. The **Detector** focuses on identifying abnormal behaviour, the **Analyst** performs contextual diagnosis, the **Responder** prepares and controls remediation actions, and the **Reporter** converts the final incident timeline into structured postmortem knowledge.

The project is intended as an engineering and research prototype rather than a claim of unrestricted autonomous operations. Human approval remains part of the architecture for consequential production actions. The central research direction is to evaluate whether explicit agent specialisation, structured handoffs and historical context can improve incident-response quality compared with a single general-purpose agent.

1.2 Target Beneficiary

The primary beneficiaries are DevOps engineers, site reliability engineers, cloud engineers, platform teams and technical operations teams responsible for maintaining production services. Engineering managers and project evaluators can also use the system to inspect incident timelines, proposed actions, outcomes and generated postmortems.

- **DevOps/SRE engineers:** reduced repetitive investigation and documentation effort.
- **Cloud and platform engineers:** structured diagnosis and controlled remediation recommendations.
- **Operations teams:** consolidated incident context and a consistent response workflow.
- **Engineering managers:** visibility into incident timelines, response quality and recurring failure patterns.
- **Academic researchers:** an experimental platform for evaluating multi-agent specialisation and context propagation.

1.3 Project Scope

The project will implement a working prototype for cloud incident management using synthetic or controlled operational incidents. The scope covers detection, diagnosis, remediation recommendation, human approval, resolution verification and postmortem generation.

In Scope	Out of Scope
Cloud/DevOps incident detection using metrics and logs	Universal support for every cloud provider and monitoring platform
Specialised Detector, Analyst, Responder and Reporter agents	Unrestricted autonomous control of production infrastructure
Event-driven communication between agents	Replacing established commercial AIOps platforms

LLM-assisted diagnosis and structured reasoning	Guaranteeing perfect root-cause accuracy
Risk-based remediation proposal and human approval	Autonomous high-risk production changes without approval
Incident timeline, postmortem and historical knowledge	Legal, operational or security certification
Controlled evaluation against incident scenarios	Claiming superiority over proprietary commercial systems without experiments

Project Objectives:

- Design a modular multi-agent architecture for the end-to-end incident lifecycle.
- Detect operational anomalies and convert them into structured incident events.
- Provide context-rich root-cause analysis using metrics, logs, deployments and historical incidents.
- Generate ranked remediation proposals with explicit risk and approval states.
- Maintain human authority over consequential infrastructure actions.
- Generate structured postmortems and preserve incident knowledge for future diagnosis.
- Evaluate specialised-agent and general-agent approaches using measurable quality metrics.

2 PROJECT DESCRIPTION

2.1 Reference Algorithm

The reference method is an event-driven incident-management algorithm. It separates detection from reasoning and reasoning from infrastructure authority.

```
MultiAgentIncidentResponse(incident):
    event <- Detector.detect(incident.telemetry)
    context <- Analyst.collectContext(event)
    diagnosis <- Analyst.reason(context)
    actions <- Responder.propose(diagnosis)
    decision <- ApprovalPolicy.evaluate(actions)
    if decision.requiresHumanApproval:
        decision <- HumanApproval.review(actions)
    if decision.approved:
        result <- Responder.execute(actions)
        verify(result)
    report <- Reporter.generateTimeline(event, context, diagnosis, actions, result)
    KnowledgeBase.store(report)
    return report
```

Incident State Machine

State	Meaning
DETECTED	An anomaly has crossed the incident-detection boundary.
INVESTIGATING	The Analyst is collecting and correlating evidence.
DIAGNOSED	A probable cause and supporting evidence have been produced.
ACTION_PROPOSED	The Responder has generated one or more remediation options.
AWAITING_APPROVAL	A consequential action requires human confirmation.
REMEDIATING	An approved action is being executed.
RESOLVED	The service has recovered or the incident is otherwise contained.
DOCUMENTED	A postmortem and action items have been generated.
CLOSED	Incident workflow completed and indexed into organizational memory.

2.2 Data Structures

Structure	Purpose	Representative Fields
Incident Record	Canonical representation of incident	incident_id, service, severity, status, start_time, end_time
Telemetry Event	Raw or normalized metrics	metric, value, timestamp, source, threshold, anomaly_score
Log Evidence	Application & system logs	log_id, timestamp, service, message, pattern, severity
Diagnosis Record	Analyst output	root_cause, confidence, evidence_ids, blast_radius, alternatives
Remediation Action	Candidate or executed response	action_id, type, risk, rationale, approval_status, result
Postmortem	Structured incident knowledge	impact, root_cause, timeline, remediation, action_items

3 SYSTEM REQUIREMENTS

3.1 User Interface & Agent Architecture

The web dashboard provides an operational control view displaying live telemetry, agent execution status, pending approvals, and historical incident postmortems.

4 NON-FUNCTIONAL REQUIREMENTS

4.1 Performance Requirements

Requirement	Target Target	Measurement Condition
Incident event creation	≤ 1 second	Local event generation after detector decision
Event delivery	≤ 500 ms typical	Healthy event bus and local deployment
Dashboard response	≤ 2 seconds	Indexed query without external cloud call
Historical incident retrieval	≤ 2 seconds	Typical similarity/search request
Rule/risk evaluation	≤ 500 ms	Local deterministic policy checks

4.2 Security Requirements

Control Area	Requirement
Authentication	Centralized authentication with MFA support where available.
Authorization & RBAC	Role-based access control for operator, administrator, and auditor roles.
LLM Input Safety	Treat logs and telemetry as untrusted data; isolate instructions from evidence to prevent prompt injection.
Remediation Safety	Block high-risk infrastructure execution until explicit human approval is recorded.

5 OTHER REQUIREMENTS & RESEARCH EVALUATION

5.1 Controlled Incident Scenarios

Scenario	Expected Behaviour	Acceptance Evidence
CPU saturation	Detector identifies anomaly; Analyst attributes likely resource saturation.	Incident event, diagnosis and confidence score.
Application error spike	Detector raises incident; Analyst correlates logs and recent deployment changes.	Relevant evidence references and root-cause candidate.
Database connection exhaustion	Analyst identifies database pool evidence and proposes safe remediation.	Diagnosis, action proposal and approval record.
Failed deployment	Analyst correlates deployment event with service degradation.	Deployment evidence and diagnosis.
High-risk remediation	Responder pauses for human approval gate.	Approval/rejection event and execution record.

5.2 Empirical Research Evaluation Model

Evaluation Dimension	Definition	Evidence / Metric
Detection Accuracy	Proportion of known incidents detected without missed alerts.	Incident dataset & detector output
Diagnosis Accuracy	Whether correct root cause is identified or ranked highly.	Ground-truth scenario labels
Diagnosis Latency	Time between incident event and structured diagnosis.	Timestamped agent events
Context Efficiency	Information transferred relative to diagnostic benefit.	Payload size, token usage & accuracy
Safety Compliance	Ability to prevent unauthorized consequential actions.	Approval and execution audit records